ITI1120 – Programming I
Fall 2024
Course Information Sheet

Contact Information:	Instructor:	Monica Vahdati
	Section #:	F
	Office:	SITE building Room 5-077
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Prerequisite(s): None

Credit Value: Three (3)

Hours per week: 3

Class Location: Wednesday 7:00 – 9:50 pm
Room: 100 Louis Pasteur (CRX) C140

Office hours: Every Wednesday from 5 to 6 PM
Please send an email that you are coming so things can be organized.
because the number of students registered is very high.

Textbook: "Introduction to Computing Using Python: An Application Development Focus, 2nd Edition" by Ljubomir Perkovic (**Highly recommended**)

Resources provided on D2L

"Practical Programming (2nd ed): An Introduction to Computer Science Using Python 3" by Paul Gries, Jennifer Campbell, Jason Montojo

"Think Python, How to Think Like a Computer Scientist" by Allen Downey (**Free**)

Course Description:

This course is designed to introduce students to the logic involved in the computing processes. They will learn the ability to develop algorithms to describe the solution to a given problem, with the implementation of a scripting language. This course uses problem solving techniques to aid students in developing solutions to computer support related problems. The intent of this course is for students to become familiar with problem solving and programming logic, and to practice that logic using a scripting language.

Major Topics:

1.0 Introduction to Computer Programs
2.0 Introduction to the Software Development Process
3.0 Using Data and Variables
4.0 Using Program Logic
5.0 Procedures
6.0 Linear Data Structures

Course Evaluation Breakdown:

ITEM	DATE (SUBJECT TO CHANGE)	VALUE
Labs	Weekly Labs	**5%**
Quizzes	Quizzes	**5%**
Assignments	6 Assignments	**21%**
Midterm	Sunday, November 3rd at 9 am – 10:30 am	**24%**
Final	Scheduled during final exams week	**45%**

Detailed Grading Scheme:

Labs (5%): We will have 10 labs, each one worth 0.5% of the final grade. You will receive 0.5% if you attend the lab and successfully complete the lab exercise (you can work on the exercise before attending the lab, but you have to demonstrate it to the TA inside the Lab). Labs will be given in person on campus.

Assignments (21%): We will have 4 assignments; each assignment is worth 5.25% of the final grade.

Quizzes (5%): We will do quizzes (scheduled and/or popup) throughout the term in lectures. The total weight of all quizzes will be 5% of the final grade. Since some students will inevitably miss some lectures due to exceptional and excusable circumstances, the lowest one or two (to be decided later) grades you receive on quizzes will be disregarded in the final calculation to all students.

Midterm examination (24%): Sunday, November 3rd at 9 am – 10:30 am

Final examination (45%): Date to be set by faculty.

Course Delivery Schedule (Subject to Change):

WEEK	ASSIGNMENT
1	Numbers, Variables, Use of Functions
2	Boolean Expressions, Strings, Function Design
3	If Statements, String Method
4	Strings, Lists
5	Memory, while loops
6	Big Oh, Memory, Global Variable
7	Swapping Algorithms, 2D lists, Shallow deep copy
8	Reading week
9	Linear Search, Binary Search, Selection Sort
10	Sets and Dictionaries, Objects I
11	Objects II, Objects III (inheritance)
12	Recursion I, Recursion II
13	Recursion III + Review

Required Software

- Python IDLE (available for free on-line if you want to install at home)
- Colab (https://colab.research.google.com/)

Course Policy:

Attendance:
"Please note that, in accordance with our faculty regulations, attendance during lectures, laboratory periods and discussion groups is mandatory. **To meet accreditation requirements, to be admitted to the final examination in a subject, a student must have attended a minimum of 80% of classes and must not have more than five unauthorized or unjustified absences**."

Missed assignments: mark of **zero**.

Late assignments: -20% for every day your submission is late (if you submit between 1 minute and 24 hours late, it is considered one day), mark of zero after 2 days late

Absence from final examination: see faculty of engineering regulations.